Note:

i. All questions are compulsory.
ii. Figures to the right indicate full marks.
iii. Graph papers are not necessary for L.P.P. Only rough sketch of the graph is expected.
iv. Answers to both the sections should be written in the separate answer books.
v. Answer to every new question must be written on a new page.

Section – I

Q.1. Attempt any SIX of the following: **[12]**

i. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 5 & 4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 & 3 \\ -2 & 5 & 0 \end{bmatrix}$

then find $2A + 3B$. (2)

ii. If the function f is continuous at $x = 1$, then find $f(1)$.

Where $f(x) = \dfrac{x^2 - 3x + 2}{x - 1}$ for $x \neq 1$. (2)

iii. If $x = \tan^{-1} t$ and $y = t^3$, find $\dfrac{dy}{dx}$. (2)

iv. Evaluate: $\int \sin^2 x \, dx$. (2)

v. Write negation of the following statements:

a. Chetan has black hair and blue eyes.

b. $\exists x \in R$ such that $x^2 + 3 > 0$. (2)

vi. If $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then find $|AB|$. (2)

vii. Evaluate: $\int \dfrac{dx}{4 - 9x^2}$ (2)

viii. If the function f is continuous at $x = 2$, then find 'k'

where $f(x) = \dfrac{x^2 + 5}{x - 1}$, for $1 < x \leq 2$

$= kx + 1$, for $x > 2$ (2)

Q.2. (A) Attempt any TWO of the following: **[6][14]**

i. If $x^y = e^{x-y}$, show that $\dfrac{dy}{dx} = \dfrac{\log x}{(1 + \log x)^2}$ (3)

ii. If $\sin y = x \sin(a + y)$

prove that $\dfrac{dy}{dx} = \dfrac{\sin^2(a + y)}{\sin a}$ (3)

iii. Discuss extreme values of the function $f(x) = x \log x$. (3)

(B) Attempt any TWO of the following:

i. Discuss the continuity of the function f at $x = 0$,

where $f(x) = \dfrac{5^x + 5^{-x} - 2}{\cos 2x - \cos 6x}$, for $x \neq 0$

$= \dfrac{1}{8}(\log 5)^2$, for $x = 0$ (4)

ii. The expenditure E_C of a person with income I is given by $E_C = (0.000035)\, I^2 + (0.045)\, I$
Find marginal propensity to consume (MPC) and average propensity to consume (APC) when I = 5000. (4)

iii. Evaluate: $\int x \cot^{-1} x \, dx$ (4)

Q.3. (A) Attempt any TWO of the following: **[6][14]**

i. If p : It is a day time.
q : It is warm.
Given the verbal statements for the following symbolic statements:

a. $p \wedge \sim q$

b. $p \vee q$

c. $p \leftrightarrow q$ (3)

ii. Using the truth table, examine whether the statement pattern $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$ is a tautology, a contradiction or a contingency. (3)

iii. The cost C of producing x articles is given as
$C = x^3 - 16x^2 + 47x$.
For what values of x will the average cost be decreasing? (3)

(B) Attempt any TWO of the following:

i. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix}$ then find A^{-1} by using elementary transformation. (4)

ii. Evaluate: $\int_0^3 \dfrac{dx}{x + \sqrt{9 - x^2}}$ (4)

iii. Find the volume of a solid obtained by the complete revolution of the ellipse $\dfrac{x^2}{36} + \dfrac{y^2}{25} = 1$ about X - axis. (4)